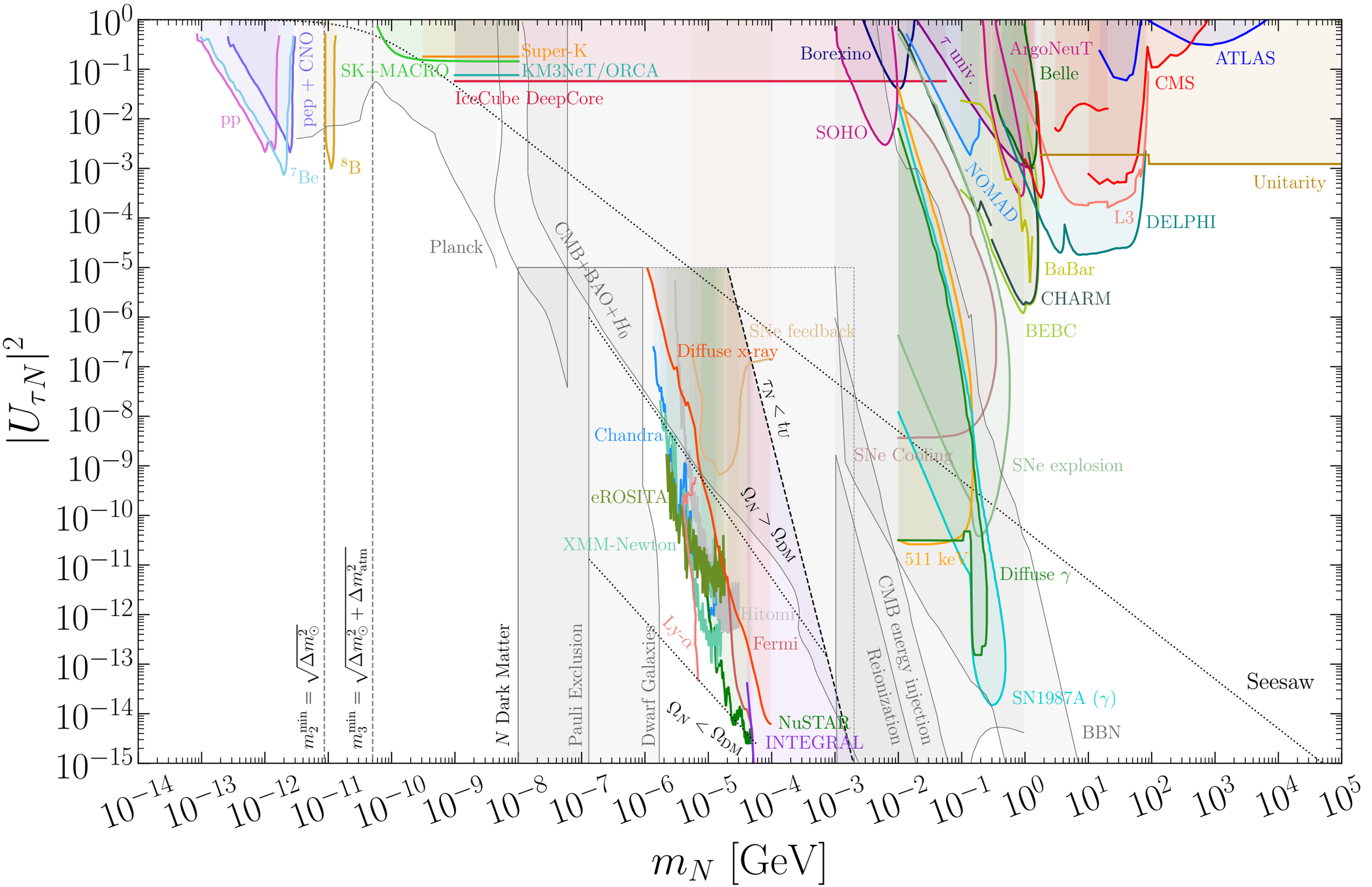

$|U_{\tau N}|^2$
m_N [GeV]
pp
^{7}Be
pep + CNO
^{8}B
SK+MACRO
Super-K
KM3NeT/ORCA
IceCube DeepCore
Borexino
SOHO
τ univ.
ArgoNeuT
Belle
ATLAS
CMS
Unitarity
NOMAD
L3
DELPHI
BaBar
CHARM
BEBC
Planck
CMB+BAO+H_0
SNe feedback
Diffuse x-ray
$\tau_N < t_U$
Chandra
eROSITA
XMM-Newton
$\Omega_N > \Omega_{DM}$
SNe Cooling
SNe explosion
511 keV
Diffuse γ
Hitomi
Ly-α
Fermi
$\Omega_N < \Omega_{DM}$
NuSTAR
INTEGRAL
SN1987A (γ)
BBN
Seesaw
N Dark Matter
Pauli Exclusion
Dwarf Galaxies
CMB energy injection
Reionization
$m_2^{\rm min} = \sqrt{\Delta m_\odot^2}$
$m_3^{\rm min} = \sqrt{\Delta m_\odot^2 + \Delta m_{\rm atm}^2}$